1 $\boldsymbol{a}=\begin{pmatrix}2\\3\end{pmatrix}$ and $\boldsymbol{b}=\begin{pmatrix}1\\5\end{pmatrix}$

(a) Write down as a column vector

(i) **a** + **b** $\begin{pmatrix}2\\3\end{pmatrix}+\begin{pmatrix}1\\5\end{pmatrix}$

$\begin{pmatrix}3\\8\end{pmatrix}$

(1)

(ii) 2**a** + 3**b** $2\begin{pmatrix}2\\3\end{pmatrix}+3\begin{pmatrix}1\\5\end{pmatrix}$

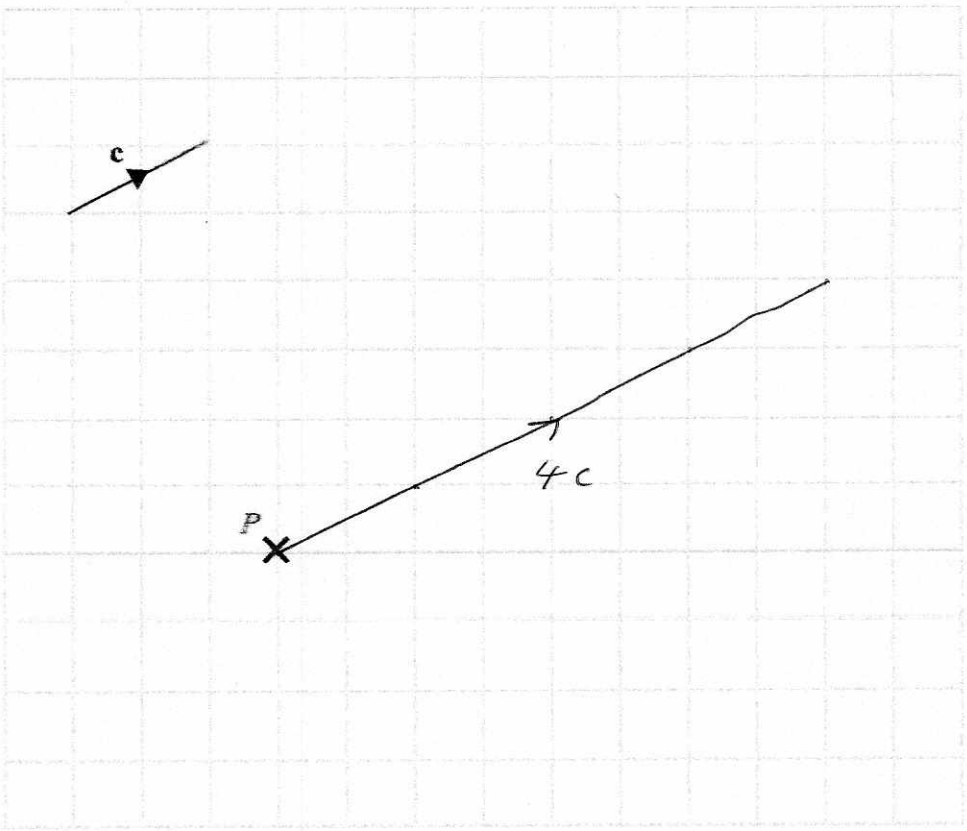
$\begin{pmatrix}4\\6\end{pmatrix}+\begin{pmatrix}3\\15\end{pmatrix}$

$\begin{pmatrix}7\\21\end{pmatrix}$

(2)

The vector **c** is drawn on the grid.

(b) From the point *P*, draw the vector 4**c**



(1)

(Total for question 1 is 4 marks)

2 $\boldsymbol{a}=\begin{pmatrix}4\\1\end{pmatrix}$ and $\boldsymbol{b}=\begin{pmatrix}3\\2\end{pmatrix}$

(a) Write down as a column vector

(i) **a** + **b** $\begin{pmatrix}4\\1\end{pmatrix}+\begin{pmatrix}3\\2\end{pmatrix}$

$\begin{pmatrix}7\\3\end{pmatrix}$

(1)

(ii) 2**a** – **b** $2\begin{pmatrix}4\\1\end{pmatrix}-\begin{pmatrix}3\\2\end{pmatrix}$

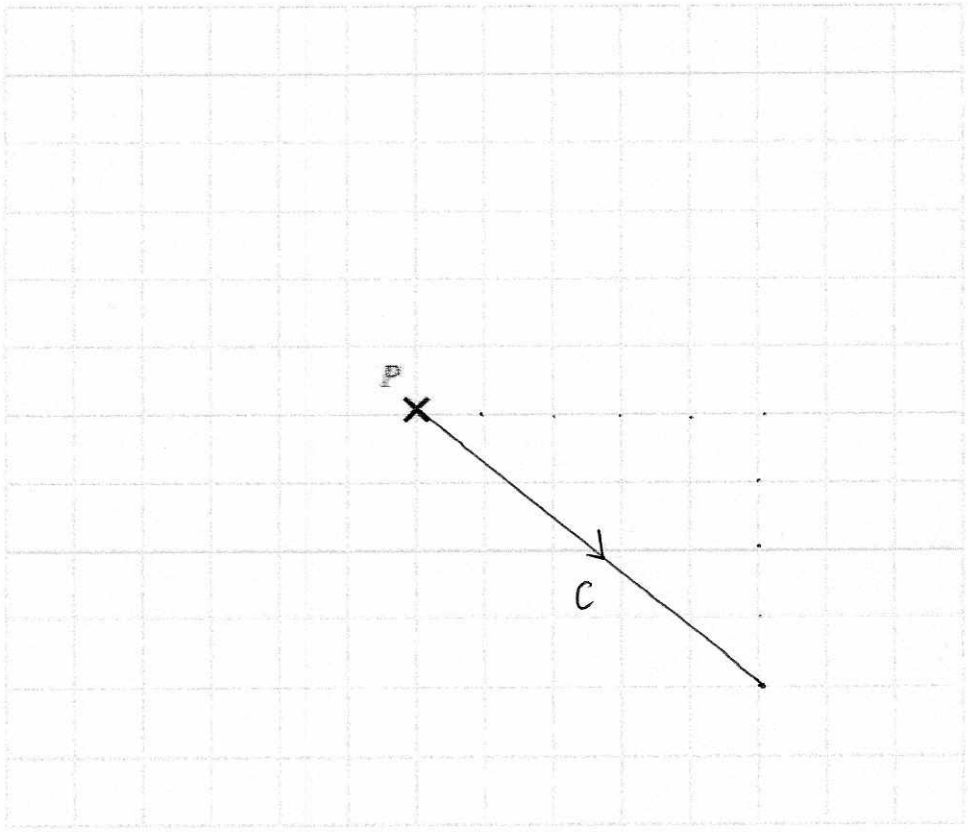
$\begin{pmatrix}8\\2\end{pmatrix}-\begin{pmatrix}3\\2\end{pmatrix}$

$\begin{pmatrix}5\\0\end{pmatrix}$

(2)

$\boldsymbol{c}=\begin{pmatrix}5\\-4\end{pmatrix}$

(b) From the point *P*, draw the vector **c**



(1)

(Total for question 2 is 4 marks)

3 $\boldsymbol{a}=\begin{pmatrix}-2\\3\end{pmatrix}$ and $\boldsymbol{b}=\begin{pmatrix}5\\-1\end{pmatrix}$

(a) Write down as a column vector

(i) **a** + **b**

$\begin{pmatrix}-2\\3\end{pmatrix}+\begin{pmatrix}5\\-1\end{pmatrix}$

$\begin{pmatrix}3\\2\end{pmatrix}$

(1)

(ii) 2**a** – **b**

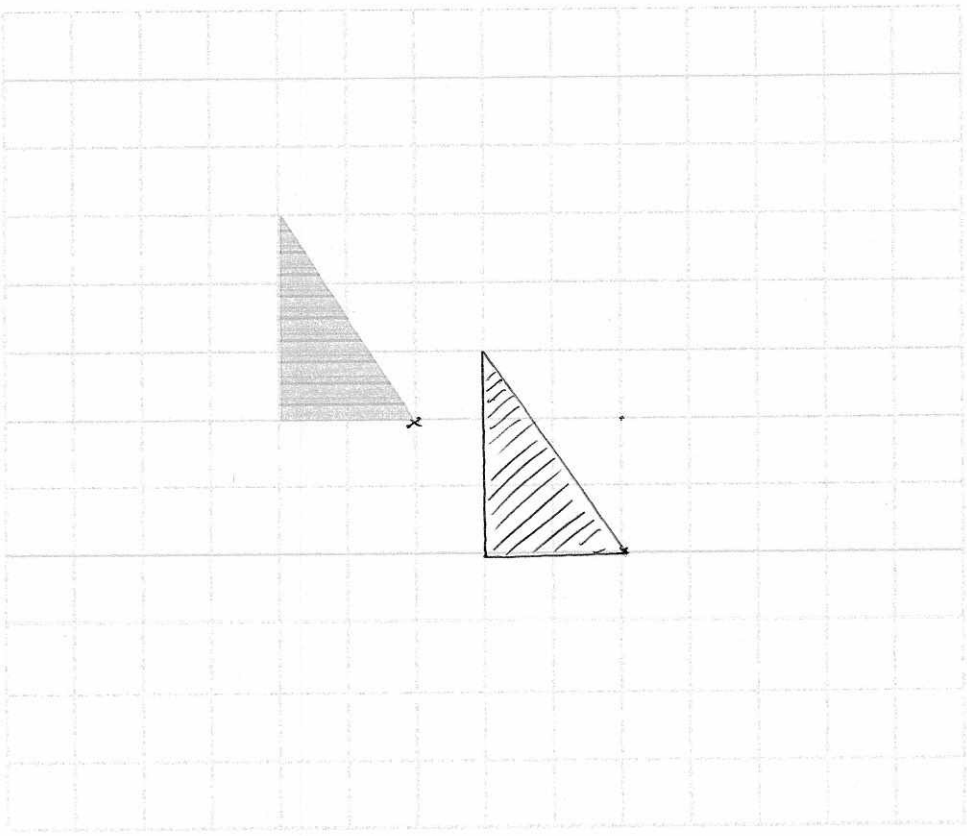
$2\begin{pmatrix}-2\\3\end{pmatrix}-\begin{pmatrix}5\\-1\end{pmatrix}$

$\begin{pmatrix}-4\\6\end{pmatrix}-\begin{pmatrix}5\\-1\end{pmatrix}$

$\begin{pmatrix}-9\\7\end{pmatrix}$

(2)

(b) Translate the triangle by the vector $\begin{pmatrix}3\\-2\end{pmatrix}$



(1)

(Total for question 3 is 4 marks)

4 A is the point (3, 2) and B is the point (4, –1).

(a) Write down as a column vector $\overrightarrow{AB}$

$$\begin{pmatrix} 4 \\ -1 \end{pmatrix} - \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

$\begin{pmatrix} 1 \\ -3 \end{pmatrix}$

(1)

C is the point (5, –2) and D is the point (2, 1).

(b) Write down as a column vector $\overrightarrow{CD}$

$$\begin{pmatrix} 2 \\ 1 \end{pmatrix} - \begin{pmatrix} 5 \\ -2 \end{pmatrix}$$

$\begin{pmatrix} -3 \\ 3 \end{pmatrix}$

(1)

(Total for question 4 is 2 marks)

5 A is the point (5, –1) and B is the point (4, –3).

(a) Write down as a column vector $\overrightarrow{AB}$

$$\begin{pmatrix} 4 \\ -3 \end{pmatrix} - \begin{pmatrix} 5 \\ -1 \end{pmatrix}$$

$\begin{pmatrix} -1 \\ -2 \end{pmatrix}$

(1)

C is the point (1, 6) and D is the point (–3, 9).

(b) Write down as a column vector $\overrightarrow{CD}$

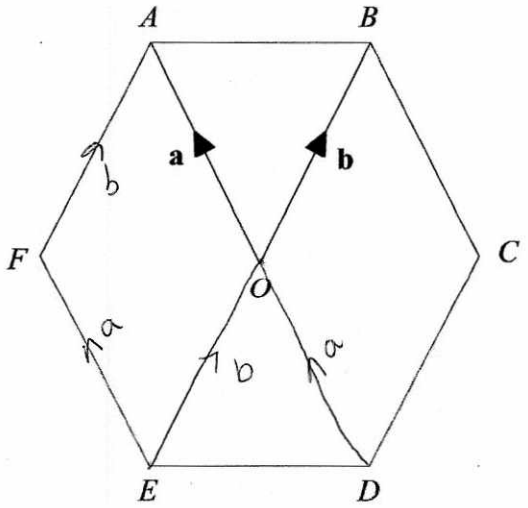
$$\begin{pmatrix} -3 \\ 9 \end{pmatrix} - \begin{pmatrix} 1 \\ 6 \end{pmatrix}$$

$\begin{pmatrix} -4 \\ 3 \end{pmatrix}$

(1)

(Total for question 5 is 2 marks)

6 *ABCDEF* is a regular hexagon with centre *O*.



$\overrightarrow{OA} = a$

$\overrightarrow{OB} = b$

(a) Find, in terms of a, the vector $\overrightarrow{AD}$

$-2a$

(1)

(b) Find, in terms of a and b, the vector $\overrightarrow{AB}$

$-a + b$

(1)

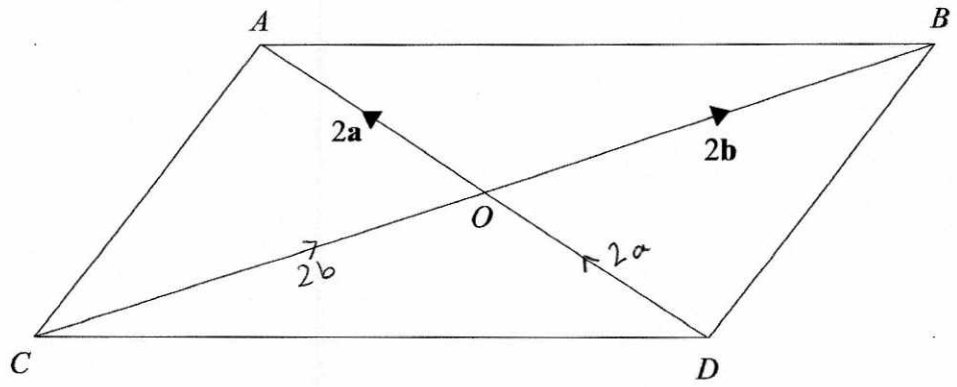
(c) Find, in terms of a and b, the vector $\overrightarrow{AF}$

$-b$

(1)

(Total for question 6 is 3 marks)

7 The diagram shows a parallelogram.



$\overrightarrow{OA} = 2\mathbf{a}$

$\overrightarrow{OB} = 2\mathbf{b}$

(a) Find, in terms of a, the vector $\overrightarrow{DA}$

4a

(1)

(b) Find, in terms of a and b, the vector $\overrightarrow{AB}$

$-2a + 2b$

(1)

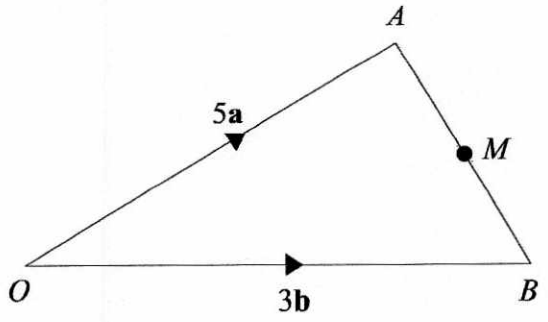
(c) Find, in terms of a and b, the vector $\overrightarrow{AC}$

$-2a - 2b$

(1)

(Total for question 7 is 3 marks)

8



$\overrightarrow{OA} = 5\text{a}$

$\overrightarrow{OB} = 3\text{b}$

M is the midpoint of AB

(a) Find, in terms of a and b, the vector $\overrightarrow{AB}$

$-5a + 3b$

(1)

(b) Find, in terms of a and b, the vector $\overrightarrow{AM}$

half of $\overrightarrow{AB}$

$-\frac{5}{2}a + \frac{3}{2}b$

(1)

(c) Find, in terms of a and b, the vector $\overrightarrow{OM}$

$5a - \frac{5}{2}a + \frac{3}{2}b$

or $5a - 2.5a + 1.5b$

$\frac{5}{2}a + \frac{3}{2}b$

(1)

(Total for question 8 is 3 marks)